Tutorial 9

The core exercises are considered part of the course material, and you are advised to at least finish these. For additional practice and interesting applications, advanced exercises are available as well – feel free to pick and choose from these.

Unless otherwise specified, we expect you to create your own algorithms for the exercises in each tutorial. This means that you should not use libraries (e.g. SciPy) for the methods that are part of the course material.

Core exercises

1. **Detecting Outliers**
   In this simple exercise we are going to consider five datasets which we will model using a linear relation.
   (a) Download the five datasets with `wget` from:

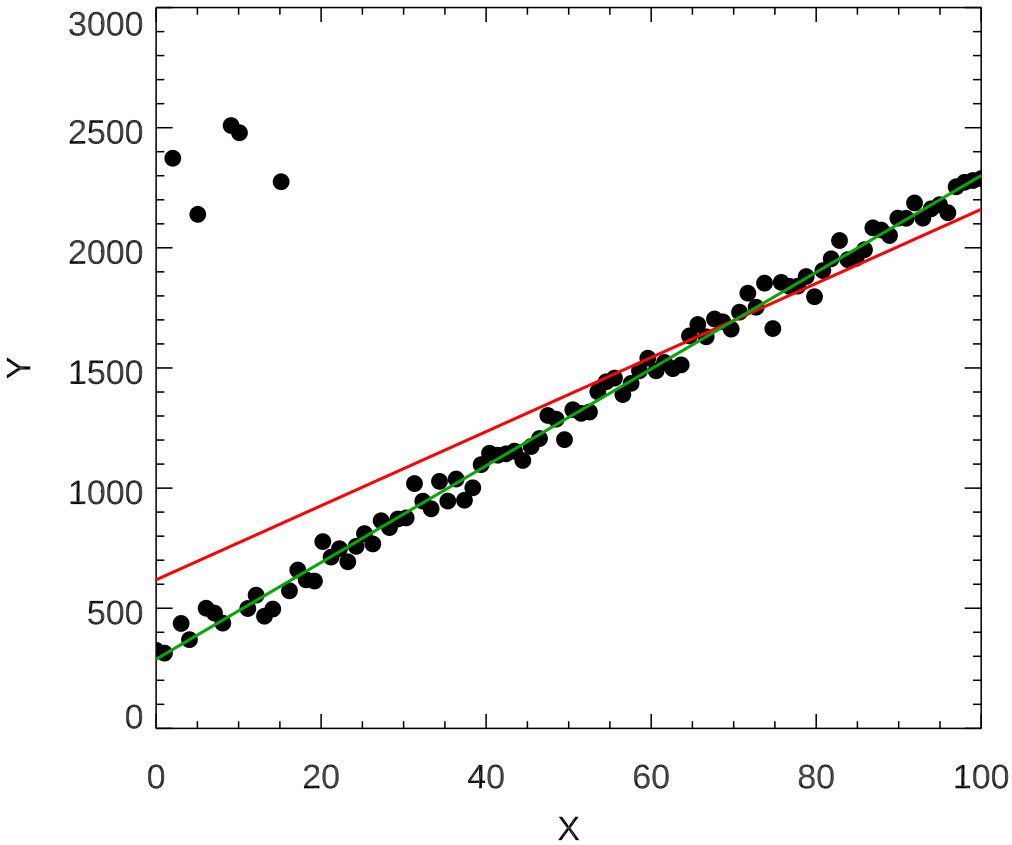
   http://home.strw.leidenuniv.nl/∼marinichenko/NUR_A/dataset<i>.txt where $< i >= 1..5$, and inspect their content with a plot.

   (b) Pick one of the dataset to your liking, fit a linear relation using least squares and compare the result to the data set.

   (c) As it can be seen , the fit does not properly reproduce the main trend of the data. Implement a robust estimate to minimize the impact of the outliers on the best fit parameters. Plot the result.

   (d) *(Optional)* Repeat the same analysis for the other four datasets.

Solution: The data contained in dataset 1 should correspond to the filled black circles in the plot below. The intrinsic model used to generate the points in all data files is $y = 274 + 20x$. Outliers are clearly seen at around $x \approx 10, y \approx 2300$



Clearly, a simple χ^2 fit is unable to properly recover the *true* model parameters (red line). The green line, instead, corresponds to the result obtained applying a robust estimate fit, using a Lorentzian optimizer. We created a function returning the cumulative sum of the natural log of the Lorentzian function (whose argument is the difference between the observed y_i and those predicted by the model at the observed x_i), and applied a downhill simplex method to search for the parameters a and b that minimize the function. We find $a = 20.26$ and $b = 274.13$, close enough to the intrinsic values.

2. **Kolmogorov-Smirnov test**
In this exercise we are going to explore how the Kolmogorov-Smirnov test works. To setup, download the set of 12 random numbers from

http://home.strw.leidenuniv.nl/~marinichenko/NUR_A/RandomNumbers<i>.txt, where `<i>` indicates the number of the dataset.

   (a) Generate a Kolmogorov-Smirnov (KS) test function.

   (b) Write a function to search which of the random number dataset/s is consistent with a Cauchy distribution.

   (c) Repeat your exercise but only use the first 100 numbers in each dataset.

   (d) Write a function to determine using the KS test if there are pairs of datasets with very similar distributions. Do this with all the numbers you have in each set, and with 100 random numbers.

   (e) *(Optional)* Instead of doing the KS test, it is almost always better to generalize the KS test to include the absolute size of the most positive and the most negative difference between the two

cumulative distribution functions. This is called the Kuiper's test. Apply the Kuiper's test to compare the distributions you downloaded[1]. Can we rule out more distributions that are not equivalent to each other?

Solution:

(a) The K-S test consists in finding the maximum of the difference between the cumulative distribution function (CDF) of a statistical reference distribution and the CDF of the data. The CDF of the reference distribution can be obtained by just integrating the associated probability distribution function (PDF). Instead, to construct the CDF associated to the data you'll first have to sort your data x_i in ascending order. Then, at each value of the sorted x_i array, the CDF will be $\mathrm{CDF}_j = \sum n_i(x_i \leq x_j)/n_{\mathrm{tot}}$, with $n_i(x_i \leq x_j)$ the number of elements in the sorted array whose values are $\leq x_j$, and n_{tot} the size of the data array (`cumsum` can be helpful here).

(b) The dataset consistent with being originated from a Cauchy distribution are number 1, 3 and 8. However, dataset 8 is the result of offsetting by 1 after multiplying by 2 the values extracted from a standard Cauchy distribution.

(c) With a reduced number of elements, our statistical test is less robust. For instance, the Q value for dataset 10 goes from $Q_{\mathrm{full}} \sim 10^{-65}$ for the full set to $Q_{100} \sim 0.1$ when the test is run on the first 100 elements only. Because $Q = 0.1$ is a value adopted as a threshold to distinguish between accepting and rejecting a null hypothesis, this result could suggest that also dataset 10 is consistent with a Cauchy distribution. In reality this is not the case, and this example stresses on the importance of significantly probing the range of values of your dataset (whenever possible).

(d) When all elements are used, the KS test should identify only dataset 1 and 3 as being extracted from the same distribution. If we reduce the number of elements to 100, dataset 1, 3, 5 and 10 appear to be similar. However, dataset 5 was generated from a Normal distribution, while dataset 10 from a Laplace distribution. Furthermore, dataset 8 and 11 are also identified to have similar distributions, while in reality dataset 11 was generated from another Laplace distribution.

3. **The chi-square test**

In this problem, we will use Pearson's chi-square χ^2 statistics to test the goodness of a fit. The chi-square statistic can be expressed in the following form for discrete data points:

$$\chi^2 = \sum_{i=1}^{\nu} \frac{(x_i - \mu)^2}{\sigma_i^2} \tag{1}$$

where μ is the mean and σ_i^2 is variance.

(a) Five different groups at CERN have measured the mass of a new hypothetical particle. Download their measurements using `wget` from:

http://home.strw.leidenuniv.nl/∼marinichenko/NUR_A/Chi-square.txt.

The mass measurement of each group is given in no particular order. Implement a chi-square test function, and apply it to the data. Consider the reduced chi-squared statistic. What does the value imply here?

Hint:

$$\mu = \frac{\sum_i m_i/\sigma_i^2}{\sum_i 1/\sigma_i^2}, \tag{2}$$

$$\sigma_\mu^2 = \frac{1}{\sum_i 1/\sigma_i^2}. \tag{3}$$

[1] For more information on the Kuiper's test see Chapter 14.3.4 in Numerical Recipes

(b) To check the significance of the goodness of fit (we fit with a constant value) in the previous part, calculate the P-value, i.e. the probability of finding a value higher than the χ^2 obtained above, given that the probability density function of the χ^2-distribution has the form:

$$f(\chi^2) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} e^{-\chi^2/2} (\chi^2)^{(-1+\nu/2)}. \tag{4}$$

Use the improper integration routine you have created in Tutorial 4 to estimate the value of the integral.

(c) *(Optional)* Download a new dataset from:
http://home.strw.leidenuniv.nl/∼marinichenko/NUR_A/Chi-square2.txt
and check the goodness of fit assuming a linear model $y = ax + b$ with $a = 0.16$ and $b = 0.83$. Compute the reduced χ^2 statistic and the P-value. What do these values tell us?

(d) *(Optional)* Perform a G-test for the above 2 problems. The test statistic is a chi-squared statistic with the same degrees-of-freedom as the analogous chi-squared test. You can use the same distribution table. What do the statistic and P-value signify here in comparison to the previously obtained chi-squared statistic?

Solution:

(a) The reduced $\chi^2_{\rm red} \sim 2.3/4 \sim 0.57$. Being $\chi^2_{\rm red} < 1$, we could infer that the average value is consistent with our data. Beware that small values of χ^2 could be produced in those cases where the uncertainties are large (either overestimated, or because the data has poor S/N).

(b) However, the corresponding P-value for a $\chi^2 = 2.3$ and 4 degrees of freedom is $P \sim 0.68$, still significant, making our observations less likely to be consistent with a constant μ.